

Wednesday 14 October 2020 – Afternoon

A Level Mathematics A

H240/02 Pure Mathematics and Statistics

Printed Answer Booklet

Time allowed: 2 hours

You must have:

- Question Paper H240/02 (inside this document)
- a scientific or graphical calculator



Please write clearly in black ink. **Do not write in the barcodes.**

Centre number

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Candidate number

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First name(s)

Last name

INSTRUCTIONS

- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided in the **Printed Answer Booklet**. If you need extra space use the lined pages at the end of the Printed Answer Booklet. The question numbers must be clearly shown.
- Answer **all** the questions.
- Where appropriate, your answer should be supported with working. Marks might be given for using a correct method, even if your answer is wrong.
- Give non-exact numerical answers correct to 3 significant figures unless a different degree of accuracy is specified in the question.
- The acceleration due to gravity is denoted by $g \text{ m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

INFORMATION

- The total mark for this paper is **100**.
- The marks for each question are shown in brackets [].
- This document has **20** pages.

ADVICE

- Read each question carefully before you start your answer.

Section A: Pure Mathematics

1(a)(i)

$$y = (2x+3)^7$$

$$\frac{dy}{dx} = 7(2x+3)^6 \times 2$$

$$= 14x(2x+3)^6$$

1(a)(ii)

$$y = x^3 \cdot \ln x$$

$$u = x^3 \quad \dot{u} = 3x^2$$

$$v = \ln x \quad \dot{v} = \frac{1}{x}$$

$$\frac{dy}{dx} = x^2 + 3x^2 \ln(x)$$

1(b)

$$\int \cos 5x \, dx = \frac{1}{5} \sin 5x + c$$

1(c)

$$\frac{dy}{dx} = 6x - 5$$

$$y = 3x^2 - 5x + c \quad (1, 3)$$

$$3 = 3 - 5 + c$$

$$c = 5$$

$$y = 3x^2 - 5x + 5$$

2

$$\begin{array}{r} 2x+3 \\ x^2-x-2 \overline{) 2x^3+x^2-7x-6} \\ \underline{-(2x^3-2x^2-4x)} \\ 3x^2-3x-6 \\ \underline{-(3x^2-3x-6)} \\ 0 \end{array}$$

$$= \underline{\underline{2x+3}}$$

3(a)(i)

$$(1-x)^{-2} = \cancel{1+x} \dots$$

$$= 1 + \frac{(-2)(-x)}{1!} + \frac{(-2)(-3)(-x)^2}{2!} + \frac{(-2)(-3)(-4)(-x)^3}{3!}$$

$$= 1 + 2x + 3x^2 + 4x^3$$

3(a)(ii)

$$\frac{(n+1)! x^n}{n!} = \underline{\underline{(n+1)x^n}}$$

$$\frac{(n+1)!}{n!} = \frac{(n+1)(n)(n-1)(n-2) \dots \times 1}{(n)(n-1)(n-2) \dots \times 1} = n+1$$

3(b)

$$a=1, r=x$$

$$S_{\infty} = \frac{a}{1-r} = \frac{1}{\underline{\underline{1-x}}}$$

3(c)

~~1/2~~ 2

$$2 + 3x + 4x^2 + 5x^3 = 1 + x + x^2 + x^3 + \dots \quad (A)$$

$$\textcircled{+} 1 + 2x + 3x^2 + 4x^3 + \dots \quad (B)$$

$$(A) = \frac{1}{1-x}$$

$$B = (1-x)^{-2} = \frac{1}{(1-x)^2}$$

$$\therefore A + B = \frac{1}{(1-x)} + \frac{1}{(1-x)^2}$$

$$= \frac{(1-x)^2 + (1-x)}{(1-x)^3} = \frac{1-2x+x^2+1-x}{(1-x)^3}$$

$$= \frac{x^2 - 3x + 2}{(1-x)^3}$$

$$= \frac{(x-1)(x-2)}{(1-x)^3} = \frac{-(x-2)}{(1-x)^2}$$

$$= \frac{2-x}{(1-x)^2}$$

4 $\cos^2 \phi = X$

$$3X^2 + X - 4 = 0$$

$$(3X + 4)(X - 1) = 0$$

$$X = -\frac{4}{3}$$

$$\cos^2 \phi = -\frac{4}{3}$$

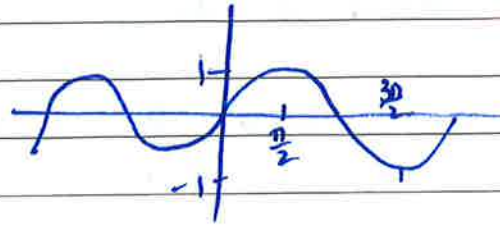
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$$X = 1$$

$$\cos^2 \phi = 1$$

$$\sin \phi = \pm 1$$

$$\therefore \phi = \pm \frac{\pi}{2}, \frac{3\pi}{2}$$



5(a)

$$\frac{n^2 - 1}{2}$$

$$\frac{n^2 + 1}{2}$$

$$n^2 - 1 = 2m \text{ (even)}$$

$$n^2 = 2m + 1$$

$\therefore n^2$ is odd,

$$n^2 + 1 = 2m \text{ (EVEN)}$$

$$n^2 = 2m - 1$$

$\therefore n^2 = \text{odd}$

$\therefore n$ is odd, but $n > 1$

5(b)

$$n^2 + \left(\frac{n^2-1}{2}\right)^2$$

$$= n^2 + \frac{n^4 - 2n^2 + 1}{4}$$

$$= \frac{4n^2}{4} + \frac{n^4 - 2n^2 + 1}{4}$$

$$= \frac{n^4 + 2n^2 + 1}{4} = \frac{(n^2 + 1)^2}{4} = \left(\frac{n^2 + 1}{2}\right)^2 \quad \text{QED}$$

6

$$\sqrt{2} \cos(2\theta + 45^\circ) = \sqrt{2} [\cos 2\theta \cos 45^\circ - \sin 2\theta \sin 45^\circ]$$

$$= \sqrt{2} \left[\frac{1}{\sqrt{2}} \cos 2\theta - \frac{1}{\sqrt{2}} \sin 2\theta \right]$$

$$= \cos 2\theta - \sin 2\theta$$

$$= (\cos^2 \theta - \sin^2 \theta) - 2 \sin \theta \cos \theta$$

$$= \cos^2 \theta - 2 \sin \theta \cos \theta - \sin^2 \theta \quad \text{QED.}$$

7(a)	length of AB
7(b)	midpoint of AB
7(c)(i)	$\frac{1}{2}(a+b)$ $\frac{1}{2}(a+b)$
7(c)(ii)	$\frac{ (a-b) }{2}$
7(d)	centre = $\frac{1}{2} \left[\begin{pmatrix} 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix} \right]$ $\leftarrow$ AB centre = midpoint of a & b = $\frac{1}{2}(a+b)$ $= \frac{1}{2} \left[\begin{pmatrix} 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix} \right] = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ $\therefore$ RADIUS = $\frac{1}{2}$ length = $\left \frac{1}{2} \left[\begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \end{pmatrix} \right] \right$ $= \left \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right = \sqrt{1^2 + 3^2} = \sqrt{10}$ $\therefore (x-3)^2 + (y-2)^2 = 10$

8(a)

$$\frac{dP}{dt} = (100 - P)$$

$$\int \frac{1}{100-P} dP = \int dt$$

$$\checkmark -\ln(100-P) = t + C \quad @ t=0, P=2000$$

$$-\ln(100-2000) = C$$

$$-\ln|-1900| = C$$

$$\checkmark C = -\ln(1900) = \ln\left(\frac{1}{1900}\right)$$

$$\therefore -\ln(100-P) = t - \ln(1900) \quad (x-1)$$

$$\Rightarrow \ln(100-P) = -t + \ln(1900)$$

$$\ln(1900) - \ln(100-P) = t$$

$$\ln\left(\frac{1900}{100-P}\right) = t \quad e^{}$$

$$\frac{1900}{100-P} = e^t$$

$$\frac{1900}{e^t} = 100-P$$

$$1900e^{-t} = 100-P$$

$$P = 100 + 1900e^{-t}$$

change to + since @ $t=0$,
 $P=2000$

8(b)

Starts at 2000, decreases exponentially to approach
and TENDS TO 100



